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# CQF Exam Three

Machine Learning in Finance

June 2026 Cohort

## Instructions

The submitted report must present work and outputs clearly separated by Question. Submit **ONLY ONE** zip file named `LASTNAME.zip` that includes your compiled PDF analytical report, clean production-grade code files (`.py` scripts), HTML reports, data tracking links, and any other supporting working files.

A raw Python notebook with auxiliary output (data plots, unformatted text) alone is **not** an analytical report; such submissions will receive a strict deduction. Please do not discuss this assignment in groups or messaging channels. Raise a support ticket for your queries. Only clarifying questions are allowed.

### Introduction

Short-term asset returns are exceptionally challenging to predict. Efficient markets produce near-normal daily returns with no significant autocorrelation between $r_t$ and $r_{t-1}$. This exam is a rigorous exercise in supervised and unsupervised machine learning. You are expected to explore a diverse and comprehensive universe of initial features of your choice, with both the original feature pool and the final selected feature subset being sufficiently numerous to justify an institutional-grade quantitative pipeline.

### Objective

Your objective is to develop a predictive framework to forecast positive market moves (uptrends) utilizing advanced ensemble gradient boosting techniques. Your proposed solution must be comprehensive, including detailed feature engineering, robust handling of time-series constraints, and structural model architecture comparisons.

- **Asset Universe:** Choose one liquid ticker of your choice from an index, individual equity, ETF, crypto token, or commodity.
- **Target Variable:** Predict the trend for short-term returns using binomial classification. The dependent variable must be labeled strictly as $Y \in \{0, 1\}$, where 1 represents a meaningful positive market move and 0 aggregates flat/downward moves.
- **Noise Handling & Labeling:** Devise your own mathematically sound approach for categorizing extremely small, near-zero returns (e.g., dropping them from the training sample or grouping them with the negative/flat returns class). The threshold choice will depend on the structural volatility of your chosen ticker.

- **Dimensions:** The number of features to include is a design choice; there is no universally recommended set of features for all assets. The length of the historical dataset is also a design choice, though training and testing over a period of up to 5 years should be sufficient for daily moves.

# A. Feature Engineering & Selection Using the Funneling Approach [30 Marks]

### Question 1

Perform feature selection for your machine learning models using a multi-step funneling process by combining techniques from Filter, Wrapper, and Embedded methods.

**(a)** Explain the feature selection funnel using the three distinct categories of feature selection methods, step by step. You must provide the exact mathematical formulation for the metric or objective used in each stage (e.g., the formula for Mutual Information score, the optimization objective of the Wrapper baseline, and the regularized loss function of the Embedded method).

**(b)** Provide a formal mathematical justification for the feature transformations used to ensure stationarity (e.g., log returns or fractional differentiation equations) and document the pruning metrics at each tier of the funnel. Justify the specific statistical or structural selection criteria used to retain or discard features at each step.

**(c)** Provide the final list of selected features that will be fed into the downstream boosting architectures.

# B. Model Building, Tuning, and Benchmarking [70 Marks]

### Question 2

Predict positive market moves (uptrends) using modern Gradient Boosting Decision Tree (GBDT) architectures.

**(a)** Build and train three distinct binary classification models using the final feature subset derived in Section A: **XGBoost, LightGBM, and CatBoost**. Formally define the mathematical optimization objective function for a GBDT architecture, explicitly highlighting how $L_1$ and $L_2$ regularizations alter the structural tree scores.

**(b)** Tune the structural hyperparameters of each estimator to obtain an optimal configuration. You must tailor your tuning strategy to respect each algorithm's unique underlying mechanics. Use mathematical notation to express how LightGBM's leaf-wise split criterion chooses a node relative to XGBoost's level-wise objective reduction. Explain how CatBoost mathematically structures its Target Statistics to prevent look-ahead bias. The choice of hyperparameter search space optimization is a design decision.

**(c)** Evaluate and compare the prediction quality of all three frameworks. Your analysis must include out-of-sample benchmarking utilizing the Area Under the Receiver Operating Characteristic (ROC-AUC) curve, Confusion Matrices (analyzing Precision-Recall tradeoffs for trading execution), and comprehensive Classification Reports. Formally write out the mathematical definitions for all performance metrics used.

Provide a definitive, data-driven recommendation on which specific algorithm is best suited for this predictive task, balancing statistical predictive power, operational stability, and risk metrics.

### Optional Portfolio Add-on

As an optional add-on to fortify your final recommendation, implement a simple execution strategy backtesting the predicted signals. Utilize performance metrics (e.g., via `QuantStats` or custom vectorized implementations) to compare the risk-adjusted returns (Sharpe ratio, Sortino ratio, and Maximum Drawdown) of the model-driven strategies against a standard baseline Buy-and-Hold strategy of your chosen asset.